Identifying Articulation Points in a Graph Using DFS

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1 Basic Definitions

Definition 1.1 (DFNum)**.** *In a Depth-First Search (DFS) on a graph* $G = (V, E)$*, each vertex* $u \in V$ *is assigned a number* $DFNum(u)$ *that indicates the order in which* u *is visited during the DFS.*

Definition 1.2 (Low)**.** *For each vertex* $u \in V$*, the value* $Low(u)$ *is defined as:*

$$Low(u) = \min\Big\{ DFNum(u),\ \min_{v \in children\ of\ u} Low(v),\ \min\{DFNum(w) \mid (u, w)\ is\ a\ back\ edge\}\Big\}.$$

This value represents the smallest DFNum reachable from u or its subtree using tree edges or back edges.

Definition 1.3 (Articulation Point)**.** *A vertex* u *(except the root of the DFS tree) is called an* ***articulation point*** *(or cut vertex) if there exists at least one child* v *of* u *in the DFS tree such that:*

$$Low(v) \geq DFNum(u).$$

Moreover, the root of the DFS tree is an articulation point if it has more than one child.

2 Algorithm for Computing DFNum and Low

The following DFS algorithm simultaneously computes DFNum, Low and identifies articulation points.

Algorithm 1: Computing DFNum and Low Using DFS

Data: A graph $G = (V, E)$ and a starting vertex s

Result: Arrays DFNum and Low for every vertex, and the set of articulation points

time $\leftarrow 0$;
foreach $u \in V$ **do**
  DFNum[u] $\leftarrow 0$;

Function `DFS`(u)**:**
  time $\leftarrow$ time $+ 1$;
  DFNum[u] $\leftarrow$ time;
  Low[u] $\leftarrow$ DFNum[u];
  foreach v *neighbor of* u **do**
    if *DFNum*[v] $= 0$ **then**
      parent[v] $\leftarrow u$;
      `DFS`(v);
      Low[u] $\leftarrow$ min(Low[u], Low[v]);
    else
      if $v \neq$ *parent*[u] **then**
        Low[u] $\leftarrow$ min(Low[u], DFNum[v]);

  if $u \neq$ *root* **then**
    foreach v *child of* u **do**
      if *Low*[v] $\geq$ *DFNum*[u] **then**
        Mark u as an articulation point;

`DFS`(s);

3 Proof of Correctness

Case 1: If the root has more than one child, removing it separates the DFS subtrees.

Case 2: If a vertex v has a child u such that there is no back edge from the subtree of u to v or any of its ancestors, then removing v disconnects u and its subtree.

In both cases, the graph decomposes into separate components, showing that v is an articulation point.

We prove the correctness of the algorithm by induction:

- **Base case:** If u is a leaf, it has no children. Then according to the algorithm:

$$\text{Low}(u) = \text{DFNum}(u),$$

  which matches the definition.

- **Induction hypothesis:** Assume that for every vertex with depth less than u, the values of Low are correctly computed.

- **Induction step:** When returning from the DFS call on vertex u, the value $\text{Low}(u)$ is computed as:

$$\text{Low}(u) = \min\Big\{\text{DFNum}(u),\ \min_{v \in \text{children of } u} \text{Low}(v),\ \min\{\text{DFNum}(w) \mid (u, w) \text{ is a back edge}\}\Big\}.$$

  By the induction hypothesis, Low for each child is correct; therefore $\text{Low}(u)$ is also correctly determined.

4 Example: Finding Articulation Points in an Undirected Graph

Consider the following undirected graph:

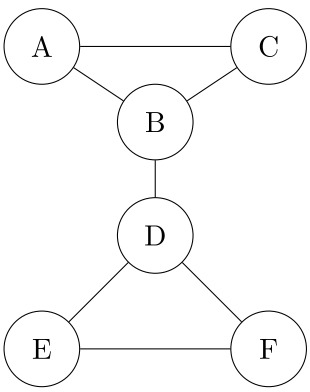

Figure 1: Undirected graph for articulation point detection

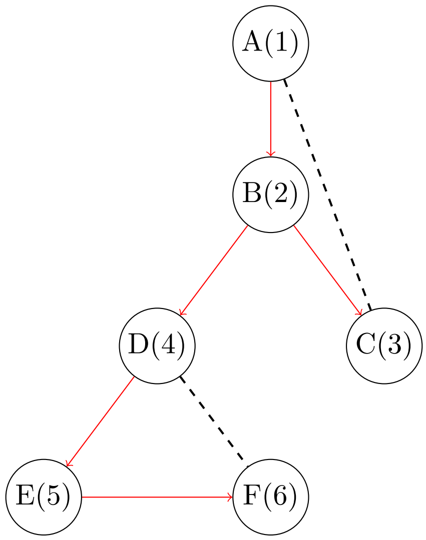

Figure 2: DFS tree of the graph

Steps of DFS and articulation point calculation

We start the DFS from vertex A. The following table shows the DFNum and Low values for each vertex.

Vertex	DFNum	Low
A	1	1
B	2	1
C	3	1
D	4	4
E	5	4
F	6	4

Table 1: DFNum and Low values for each vertex

Using the Low values and the articulation point rules:

- If a child v of u satisfies $\text{Low}[v] \geq \text{DFNum}[u]$, then u is an articulation point (except for the root).
- The root is an articulation point if it has more than one child in the DFS tree.

The set of articulation points in this graph is:

$$\{B,\ D\}$$

5 Conclusion

Using the DFS algorithm and correctly computing the values of DFNum and Low, we can identify articulation points. In the example above, removing B disconnects the subtree containing E, and removing D disconnects its child subtrees. Thus B and D are articulation points.